INCH-POUND

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PERFORMANCE SPECIFICATION

FILLER AND PRESSURIZATION UNIT, HYDRAULIC SYSTEM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. The filler and pressurization unit covered by this document will be operated in hangars, flightlines, and flight decks for servicing aircraft hydraulic systems with pressures from 0-3,000 psi.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: SA-ALC/TILDD, 485 Quentin Roosevelt Rd., Bldg 171, Kelly AFB, TX 78241-6524 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4910

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2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

ZZ-T-410 Tire, Pneumatic, Industrial

MILITARY

MIL-H-5606 Hydraulic Fluid, Petroleum Base, Aircraft, Missile, and Ordinance.

MIL-P-5515 Pump, Hydraulic, Ram, Hand Driven.

STANDARDS

FEDERAL

FED-STD-595 Colors

MILITARY

MS24322 Wheel and Hub Assemblies.

AIR FORCE - NAVY AERONAUTICAL

AN6248 Pump-Hydraulic Hand, Type 3,000

(Unless otherwise indicated, copies of the above specifications and standards are available from the Defense Automated Printing Service, Bldg 4D (DODSSP), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

AIR FORCE

65J34010 Reservoir Assembly, Fluid

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7545352 Requirements for Finishes, Protective, and Codes for SA-ALC Ground and Ground Support Equipment.

8520434 Handle, Hydraulic Filler-Pressurization Unit

(Copies of these drawings are available from SA-ALC/TILDB, Data Repository, Bldg 171, 485 Quentin Roosevelt Rd., Kelly AFB, TX 78241-6524.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. First article, this document makes provisions for first article testing (see 4.2).

3.2 Material. The unit shall be constructed of components that are corrosion resistant or treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be met in storage or in normal service. All materials of the unit shall be compatible and not affected by hydraulic fluid conforming to MIL-H-4505.

3.3 Finish. Finish shall be to the best commercial practice as supplemented by AF drawing 7545352, use finish codes as required per Table I, color number 24052 per FED-STD-595.

3.4 Components. The filler and pressurization unit shall consist of the unit being mounted on pneumatic tires, a reservoir (Air Force drawing 65J34010), hand operated high pressure pump, hand operated low pressure pump, line filter, 0-5,000 pounds per square inch (psi) pressure gage, check valve, bleed valves, servicing hose and all accessories thereto necessary for proper and efficient operation.

3.5 Design and construction. The filler and pressurization unit shall be built to withstand the strain, jars, vibrations, and other conditions incident to shipping, storage, and service. The filler and pressurization unit shall be capable of servicing the hydraulic systems of aircraft which utilize hydraulic fluid conforming to MIL-H-5606.

3.6 Performance.

3.6.1 Operation position. The filler and pressurization unit shall operate satisfactorily while in its normal plane of operation to as much as 10 degrees from horizontal plane in any direction.

3.7 Details of components.

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3.7.1 Reservoir. A hydraulic fluid reservoir, in accordance with Air Force Drawing 65J34010, shall be furnished as an integral part of the filler and pressurization unit. The reservoir material shall be fabricated from stainless steel.

3.7.2 High pressure pump. A high pressure, hand operated, double acting pump shall be installed in the system to pump hydraulic fluid from reservoir to service line. This pump shall be in accordance with Class 3000 of MIL-P-5515 and AN6248, maximum pressure 3,000 psi.

3.7.3 Low pressure pump. A low pressure, hand operated double acting pump shall be installed in the system to pump hydraulic fluid from the reservoir to service line. The pump shall be capable of delivering a minimum capacity of 2 gallons per 30-35 strokes (complete cycles). The mechanical advantage of this pump shall be such that a maximum force of 30 pounds on the pump handle (17-20 inches in length measured from the center of the pump shaft to end of the handle) shall produce the rated flow, maximum pressure 100 psi.

3.7.4 Relief valves. Pressure relief valves shall be incorporated in the unit to protect the hydraulic system of the aircraft being serviced and the filler unit. When open, the relief valves shall pass the hydraulic fluid back to the reservoir. The pressure in the low pressure system shall not exceed 100 psi and the pressure in the high pressure system shall not exceed 3300 psi. The low pressure relief valve shall be adjustable in a range of from 50 psi, or less, to 100 psi. The high pressure relief valve shall be adjustable in a range of from 100 to 3,000 psi. Each relief valve shall have a minimum proof pressure rating of 1.5 times the maximum pressure setting.

3.7.5 Check valves. A check valve or equivalent assembly shall be used to protect the low pressure system while high pressure system is in use.

3.7.6 Filter assembly. A minimum three micron filter shall be installed between pumps and service hose.

3.7.7 Pressure gage. A hydraulic pressure gage of 0-5,000 psi pressure shall be provided. The gage shall indicate fluid pressure produced by either the high or low pressure pump.

3.7.8 Hydraulic tubing and hoses. Tubing and hoses that form a part of the filler and pressurization unit shall be capable of withstanding a proof pressure of 4,500 psi and a bursting pressure of 7,500 psi.

3.7.9 Service hose assembly. The service hose assembly provided as a port of the filler and pressurization unit shall be no less than 25 feet long or more than 35 feet long.

3.7.10 Bleed valves. Bleed valves shall be provided as part of the filler and pressurization unit hydraulic system. The valves shall be used for bleeding type operations, as required, and shall return the hydraulic fluid to the reservoir.

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3.7.11 Storage compartment. A storage compartment shall be attached to the filler and pressurization unit to store fittings, service hose, and hand tools to prevent contamination of items used in servicing aircraft.

3.7.12 Mobility. The filler and pressurization unit shall be mobile at speeds not to exceed 5 mph (8 kilometers per hour) and restricted to operations on generally paved surfaces such as those associated with hangar, flightline, or flight deck operations. The ground clearance beneath the equipment at gross vehicle weight (GVW) shall be no less than 8 inches. The filler and pressurization unit shall be maneuverable by one person. Means shall be provided for adequate handling. Handholds and grasps, if used, shall conform to best commercial practice.

3.7.13 Brakes. The filler and pressurization unit shall have a parking brake to make it immobile during servicing or parked when not in use.

3.7.14 Wheels and hubs. Wheels and hubs for industrial pneumatic tires shall conform to MS24322 and withstand hangar, flightline, and flight deck environment. Three wheels shall be used as a minimum.

3.7.15 Tires. Tires used on the filler and pressurization unit shall be of type 480-8 NHS, as a minimum, in accordance with ZZ-T-410F.

3.7.16 Towbar. The filler and pressurization unit shall include a towbar using standard fitting for hook up to allow movement by a tow vehicle. If a three wheel trailer is provided, the towbar shall conform to AF Drawing 8520434 and shall be attached to the front swivel wheel.

3.8. Dimensions. The completed assembled filler and pressurization unit shall have the following dimensions as limits. Length = 60” max. (towbar retracted), width = 40” max, height = 40” max.

3.9. Safety.

3.9.1 Safety. Provisions for the safety of personnel shall be incorporated to the maximum extent possible with regards to anticipated operating conditions and environment.

3.9.2 Hazardous characteristics. Hazardous characteristics such as sharp edges, projections and hinged devices shall be avoided in the design and construction of this filler and pressurization unit.

3.9.3 Maintainability. The filler and pressurization unit shall incorporate features, to the maximum extent possible, that provide ease-of-maintenance consistent with performance, reliability, interchangeability, and cost. The use of special tools, and the need for special training shall be kept to a minimum.

4. VERIFICATION

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4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2)

b. Quality conformance inspection (see 4.3)

4.2 First article inspection. Unless otherwise specified, the first article tests shall consist of all tests described under 4.4. The contractor shall demonstrate, to the satisfaction of the procuring activity, the ability of the filler and pressurization unit to meet the requirements and pass the tests and inspections, as specified herein, before final acceptance. Failure of the first article sample to meet the requirements or pass the tests and inspections, as specified herein, shall be cause for rejection.

4.2.1 First article report. The contractor shall submit a test report in accordance with contractors format and shall cover all the tests specified in 4.4.

4.3 Quality conformance inspection.

4.3.1 Individual tests. Each filler and pressurization unit shall be subject to the following tests described under 4.4.

a. Examination of product (4.4.1)

b. Test of relief valves (4.4.5)

4.4 Tests.

4.4.1 Examination of product. Each filler and pressurization unit shall be examined for conformance to the requirements of this document.

4.4.2 Operation. The reservoir shall be filled with hydraulic fluid conforming to MIL-H-5606. The low pressure pump system shall be operated for 30-35 strokes (complete cycles) and shall pump a minimum of 2 gallons of hydraulic fluid from the completely assembled filler and pressurization unit. The high pressure pump system be operated with end of the service hose capped. The high pressure pump will produce a pressure of 3,000 psi. The mechanical advantage of each pump shall be measured during the operation tests. The mechanical advantage of the low pressure pump shall be as specified in 3.7.3 and the mechanical advantage of the high pressure pump shall be such that a force not in excess of 50-55 pounds on the pump handle (26 inches in length measured along a straight line from the pump pivot to the end of the handle) shall produce a flow of fluid at a pressure of 3,000 psi.

4.4.3 Proof pressure. A proof pressure of 4,500 psi shall be applied to the high pressure of the filler and pressurization unit at least 2 successive times, and held for 5 minutes for each pressure

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application. The high pressure relief valve shall be blocked off on the outlet side and service hose capped. The high pressure pump shall be operated to produce a stabilized pressure of 4,500 psi. There shall be no evidence of external leakage, excessive distortion or permanent set. If the pressure in the system drops below 3,800 psi within a five (5) minute period or if the check valve protecting the low pressure assembly fails, this shall be cause for rejection.

4.4.4 Pressure gage accuracy. Be accurate to one-fourth of one percent over entire range of full scale.

4.4.5 Test of relief valves. The filler and pressurization unit shall be filled with hydraulic fluid conforming to MIL-H-5606 and blocked at the end of the service hose. The high pressure relief valve and the low pressure relief valve shall be tested for conformance to 3.7.4. The relief valves shall open and pass fluid back into the reservoir when subjected to the set pressure on the valves. When the pressure falls below the set pressure, the relief valve shall close.

4.4.6 Mobility tests. Mobility tests shall be conducted to assure conformance to paragraph 3.7.12.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES (This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The filler and pressurization unit covered by this document is intended for use in hangars, flight lines, and flight line decks for servicing the hydraulic systems having a system pressure from 9 to 3,000 psi.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of the DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2).

c. Packaging requirements (see 5.1).

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6.3 Subject term (key word) listing.

Brakes
Filter
Gage
Hose
Hydraulic
Pump
Reservoir
Tires
Valves
Wheels

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Air Force - 99
Army - AL

Agent activity:
Air Force 99

Preparing activity:
Air Force - 82

Project No. 4910-0803

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1,2,3, and 8. In block 1, both the document number and revision letter should be given
2. The submitter of this form must complete blocks, 4,5,6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER **MIL-PRF-83766B**	2. DOCUMENT DATE (YYMMDD) **97/08/01**

3. DOCUMENT TITLE
FILLER AND PRESSURIZATION UNIT,HYDRAULIC SYSTEM

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite,if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. **SUBMITTER**

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (include Zip Code)	d. TELEPHONE (Include Area Code) (1) DSN (If applicable) (2)	e. DATE SUBMITTED (YYMMDD)

7. **PREPARING ACTIVITY**

a. NAME **SA-ALC/TILDD**	b. TELEPHONE (Include Area Code) (1) Commercial **(210)925-6314** (2) DSN **945-6314**
c. ADDRESS (Include Zip Code) **485 QUENTIN ROOSEVELT ROAD** **KELLY AFB TX 78241-6425**	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403 FALLS CHURCH VA 22041-3466 (703)756-2340 DSN: 289-2340

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